Group 9 - Activity 5 Progress Report

BATUMBAKAL GYM MEMBERSHIP MANAGEMENT SYSTEM

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Database Maintenance and Monitoring

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Repository: https://github.com/louisandreidg/Batumbakal-Gym

Tracker: https://docs.google.com/spreadsheets/d/1BClCa9rEs9o-D9Rp3tUZVhECipJ234zX/edit?gid=2123136180

Evidence: https://github.com/louisandreidg/Batumbakal-Gym/tree/main/db-maintenance

Summary

We reviewed the local PostgreSQL 18.6 databases from Activity 4, checked available logs, performed permitted maintenance and examined application query plans.

Project	Health result	Maintenance and priority
G01 DropBoxed	**Not reachable**	Not run - intended-account authentication unavailable; Immediate action
G02 Eventer / School Event Management	**Healthy**	ANALYZE and ordinary VACUUM on school_event_management_db.event; 2026-11-07; Follow-up action
G03 PAWPAW Veterinary	**Not reachable**	Not run - intended-account authentication unavailable; Immediate action
G04 HealthBridge Clinic	**Healthy**	ANALYZE and ordinary VACUUM on public.appointments; 2026-11-07; Follow-up action
G05 SHIORI Cafe	**Issues found**	Withheld - existing account lacks MAINTAIN/ownership; proposed commands supplied; Follow-up action
G06 ByaheRentals / Car Rental	**Healthy**	ANALYZE and ordinary VACUUM on public.vehicle; 2026-11-07; Follow-up action
G07 Applicant Management	**Healthy**	ANALYZE and ordinary VACUUM on public.department; 2026-11-07; Follow-up action
G08 Catering Management	**Issues found**	Withheld - existing account lacks MAINTAIN/ownership; proposed commands supplied; Follow-up action

Project	Health result	Maintenance and priority
G10 UniSync	**Healthy**	ANALYZE and ordinary VACUUM on public.registration; 2026-11-07; Follow-up action
G11 ReserveIT	**Healthy**	ANALYZE and ordinary VACUUM on public.reservation; 2026-11-07; Follow-up action

Healthy means the checks passed at the time of inspection and the selected maintenance completed. Dropboxed and PAWPAW could not be inspected with their intended accounts. SHIORI and Catering need maintenance permission.

SQL commands, output and full query plans are saved in each project folder. EXPLAIN was run before EXPLAIN ANALYZE, using SELECT queries only.

G01 DropBoxed

Repository: https://github.com/joshxaaaaaa/GRP1_Dropboxed

Snapshot: Earlier local ZIP; revision not established

Database/schema: dropboxed_db / dropboxed

Result: Not reachable | Priority: Immediate action

Accounts and health

Inspection: dropboxed_app; application query: dropboxed_app; maintenance candidate: dropboxed_app. Not verified this activity; prior metadata audit only.

The client returned fe_sendauth: no password supplied. This is an authentication-configuration blocker, not evidence that PostgreSQL is down.

Logs and maintenance

Current connection.txt reports a missing client password. Shared historical errors are not reliably attributable to this project unless the line identifies the account or database. The database was not inspected through an administrator workaround.

Not run - intended-account authentication unavailable

Recommendation and limits

Securely configure the original Activity 4 application-account password and finish the unresolved access-control plan; then rerun monitoring. Do not reset another project or use postgres to bypass the account.

G02 Eventer / School Event Management

Repository: https://github.com/JosephEmmanuelBabas/School_Event_Management_Web_Appli

Source: main ZIP; commit unavailable

Database/schema: grp02_event_management_db / school_event_management_db

Result: Healthy | Priority: Follow-up action

Accounts and health

Inspection: grp02_readonly (grp02_readonly_grp). Query: event_app (grp02_runtime_grp). Maintenance: event_owner (object owner).

Local snapshot: 9254 kB; 10 tables, 27 indexes, 0 views. Other sessions visible: 0; visible blocked sessions: 0. Database deadlocks counter: 0; checksum failures counter: 0. Autovacuum is on.

Selected table school_event_management_db.event: estimated live rows 0, dead rows 50, modifications since analyze 30.

Logs and maintenance

ANALYZE and ordinary VACUUM on school_event_management_db.event; 2026-11-07

ANALYZE and VACUUM completed. Related tables with inaccurate row estimates were also analyzed where permitted. The command output is saved in the project folder.

Query plan and interpretation

Source: EventRepository.java:19; JPQL countVenueConflicts translated to SQL; venue 1, today, 08:00-17:00, exclusion 0

Exact SELECT and first EXPLAIN: explain.sql/txt. Runtime evidence: plan-before.txt and plan-after.txt.

Observed plan after the last authorized action

```
Aggregate: estimated 1, actual 1.0, loops 1
Seq Scan on event: estimated 1, actual 0.0, loops 1
```

Execution time before: 2.531 ms; after: 0.054 ms.

Recommendation and limits

The conflict query used an existing index, but the event table was empty. Retest with realistic bookings before adding an index.

G03 PAWPAW Veterinary

Repository: https://github.com/rubiakylashane/GRP03_PAWPAW

Snapshot: Earlier local checkout; tested revision not established

Database/schema: group3_Pawpaw / public

Result: Not reachable | Priority: Immediate action

Accounts and health

Inspection: pawpaw_app; application query: pawpaw_app; maintenance candidate: pawpaw_app. Not verified this activity; prior metadata audit only.

The client returned fe_sendauth: no password supplied. This is an authentication-configuration blocker, not evidence that PostgreSQL is down.

Logs and maintenance

Current connection.txt reports a missing client password. Shared historical errors are not reliably attributable to this project unless the line identifies the account or database. The database was not inspected through an administrator workaround. Historical example: 2026-11-06 00:29:07 CST FATAL password authentication failed for user pawpaw_app.

Not run - intended-account authentication unavailable

Recommendation and limits

Securely configure the original Activity 4 application-account password and finish the unresolved access-control plan; then rerun monitoring. Do not reset another project or use postgres to bypass the account.

G04 HealthBridge Clinic

Repository: https://github.com/RainneQuintab/GRP03_PROJECTDB

Branch: main

Database/schema: grp03_projectdb / public

Result: Healthy | Priority: Follow-up action

Accounts and health

Inspection: hb_report (hb_readonly_grp). Query: hb_app (hb_staff). Maintenance: hb_maintenance (hb_admin).

Local snapshot: 8886 kB; 11 tables, 22 indexes, 2 views. Other sessions visible: 0; visible blocked sessions: 0. Database deadlocks counter: 0; checksum failures counter: 0. Autovacuum is on.

Selected table public.appointments: estimated live rows 2, dead rows 3, modifications since analyze 5.

Logs and maintenance

ANALYZE and ordinary VACUUM on public.appointments; 2026-11-07

ANALYZE and VACUUM completed. Related tables with inaccurate row estimates were also analyzed where permitted. The command output is saved in the project folder.

Query plan and interpretation

Source: staff_appointments.php:39; unfiltered appointment listing

Exact SELECT and first EXPLAIN: explain.sql/txt. Runtime evidence: plan-before.txt and plan-after-related.txt.

Observed plan after the last authorized action

```
Sort: estimated 2, actual 2.0, loops 1
Seq Scan on specialties: estimated 4, actual 4.0, loops 1
Seq Scan on doctors: estimated 3, actual 3.0, loops 1
Seq Scan on appointments: estimated 2, actual 2.0, loops 1
Seq Scan on patients: estimated 2, actual 2.0, loops 1
```

Execution time before: 0.301 ms; after: 0.165 ms.

Recommendation and limits

ANALYZE corrected the row estimates for appointments and related tables. Sequential scans are reasonable for this small data set. Review date filtering and existing indexes if the database grows.

G05 SHIORI Cafe

Repository: https://github.com/LeeJnsn27/SHIORIcafe-new

Branch: main

Database/schema: grp10_shioricafedb / shioricafe_schema

Result: Issues found | Priority: Follow-up action

Accounts and health

Inspection: shiori_report (shiori_readonly_grp). Query: shiori_app (shiori_runtime_grp). Maintenance: shiori_app (shiori_runtime_grp).

Local snapshot: 9894 kB; 17 tables, 49 indexes, 3 views. Other sessions visible: 0; visible blocked sessions: 0. Database deadlocks counter: 0; checksum failures counter: 0. Autovacuum is on.

Selected table shioricafe_schema.reservation: estimated live rows 1, dead rows 3, modifications since analyze 7.

Logs and maintenance

Withheld - existing account lacks MAINTAIN/ownership; proposed commands supplied

Maintenance was not run because the account lacks the required privilege. A separate maintenance role is recommended.

Query plan and interpretation

Source: get-table-occupancy.php:16; active table occupancy

Exact SELECT and first EXPLAIN: explain.sql/txt. Runtime evidence: plan-before.txt and no post-maintenance plan.

Observed plan before maintenance

```
Nested Loop: estimated 1, actual 1.0, loops 1
Seq Scan on reservation: estimated 1, actual 1.0, loops 1
Index Scan on customer: estimated 1, actual 1.0, loops 1
```

Execution time before: 0.092 ms. No maintenance comparison available.

Recommendation and limits

The plan is reasonable for the small sample. The owner should provide a maintenance role. Consider status and time indexes only if a larger booking workload shows a need.

G06 ByaheRentals / Car Rental

Repository: https://github.com/123airabels/Car_Rental_Management_System

Branch: main

Database/schema: car_rental_db / public

Result: Healthy | Priority: Follow-up action

Accounts and health

Inspection: car_report (car_readonly_grp). Query: app_runtime (car_runtime_grp). Maintenance: car_dba (car_admin_grp).

Local snapshot: 9318 kB; 17 tables, 38 indexes, 0 views. Other sessions visible: 0; visible blocked sessions: 0. Database deadlocks counter: 0; checksum failures counter: 0. Autovacuum is on.

Selected table public.vehicle: estimated live rows 8, dead rows 2, modifications since analyze 11.

Logs and maintenance

ANALYZE and ordinary VACUUM on public.vehicle; 2026-11-07

ANALYZE and VACUUM completed. Related tables with inaccurate row estimates were also analyzed where permitted. The command output is saved in the project folder.

Query plan and interpretation

Source: VehicleController.java:27; list without optional filters

Exact SELECT and first EXPLAIN: explain.sql/txt. Runtime evidence: plan-before.txt and plan-after-related.txt.

Observed plan after the last authorized action

```
Sort: estimated 8, actual 8.0, loops 1
Seq Scan on vehicle: estimated 8, actual 8.0, loops 1
Seq Scan on vehicle_category: estimated 4, actual 4.0, loops 1
Seq Scan on rental_rate: estimated 1, actual 1.0, loops 8
```

Execution time before: 0.328 ms; after: 0.203 ms.

Recommendation and limits

Updated statistics improved the estimates. If the vehicle catalog grows, compare the repeated rate lookup with a join before adding indexes.

G07 Applicant Management

Repository: https://github.com/mang2mas/grp07-applicant-management-system

Branch: main

Database/schema: grp07_applicantdb / public

Result: Healthy | Priority: Follow-up action

Accounts and health

Inspection: grp07_readonly (grp07_readonly_grp). Query: grp07_app (grp07_runtime_grp). Maintenance: grp07_owner (object owner).

Local snapshot: 8798 kB; 8 tables, 20 indexes, 0 views. Other sessions visible: 0; visible blocked sessions: 0. Database deadlocks counter: 0; checksum failures counter: 0. Autovacuum is on.

Selected table public.department: estimated live rows 5, dead rows 7, modifications since analyze 8.

Logs and maintenance

ANALYZE and ordinary VACUUM on public.department; 2026-11-07

ANALYZE and VACUUM completed. Related tables with inaccurate row estimates were also analyzed where permitted. The command output is saved in the project folder.

Query plan and interpretation

Source: index.php:20; first page without search

Exact SELECT and first EXPLAIN: explain.sql/txt. Runtime evidence: plan-before.txt and plan-after-related.txt.

Observed plan after the last authorized action

```
Limit: estimated 5, actual 5.0, loops 1
Seq Scan on position: estimated 8, actual 8.0, loops 1
Seq Scan on department: estimated 5, actual 5.0, loops 1
```

Execution time before: 0.607 ms; after: 0.169 ms.

Recommendation and limits

The current plan is suitable for the small sample. Review name-search performance with realistic data before changing indexes.

G08 Catering Management

Repository: https://github.com/yami-sumi/catering-management-system

Branch: main

Database/schema: catering_management_db / public

Result: Issues found | Priority: Follow-up action

Accounts and health

Inspection: catering_report (catering_readonly_grp). Query: catering_app (catering_runtime_grp). Maintenance: catering_app (catering_runtime_grp).

Local snapshot: 8734 kB; 12 tables, 16 indexes, 0 views. Other sessions visible: 0; visible blocked sessions: 0. Database deadlocks counter: 0; checksum failures counter: 0. Autovacuum is on.

Selected table public.package: estimated live rows 3, dead rows 5, modifications since analyze 6.

Logs and maintenance

Withheld - existing account lacks MAINTAIN/ownership; proposed commands supplied

Maintenance was not run because the account lacks the required privilege. A separate maintenance role is recommended.

Query plan and interpretation

Source: customerPortal.js:83; customer package catalog

Exact SELECT and first EXPLAIN: explain.sql/txt. Runtime evidence: plan-before.txt and no post-maintenance plan.

Observed plan before maintenance

```
Sort: estimated 270, actual 3.0, loops 1
Seq Scan on package: estimated 270, actual 3.0, loops 1
```

Execution time before: 0.106 ms. No maintenance comparison available.

Recommendation and limits

The plan estimated many more package rows than it returned. A maintenance role is needed to refresh statistics; the sample does not justify a new index.

G10 UniSync

Repository: https://github.com/CoffyJosh/unisync

Branch: main

Database/schema: grp10_unisyncdb / public

Result: Healthy | Priority: Follow-up action

Accounts and health

Inspection: unisync_auditor (unisync_readonly_grp). Query: unisync_app (unisync_runtime_grp). Maintenance: unisync_dba (unisync_admin_grp).

Local snapshot: 9102 kB; 8 tables, 21 indexes, 0 views. Other sessions visible: 0; visible blocked sessions: 0. Database deadlocks counter: 0; checksum failures counter: 0. Autovacuum is on.

Selected table public.registration: estimated live rows 946, dead rows 1, modifications since analyze 2.

Logs and maintenance

ANALYZE and ordinary VACUUM on public.registration; 2026-11-07

ANALYZE and VACUUM completed. Related tables with inaccurate row estimates were also analyzed where permitted. The command output is saved in the project folder.

Query plan and interpretation

Source: get_dashboard.php:31; account 1 and empty search

Exact SELECT and first EXPLAIN: explain.sql/txt. Runtime evidence: plan-before.txt and plan-after-related.txt.

Observed plan after the last authorized action

```
Aggregate: estimated 1, actual 1.0, loops 1
Seq Scan on organization: estimated 30, actual 30.0, loops 1
Seq Scan on event: estimated 51, actual 51.0, loops 1
Bitmap Heap Scan on registration: estimated 8, actual 9.0, loops 1
```

Execution time before: 0.408 ms; after: 0.245 ms.

Recommendation and limits

Refreshing statistics improved the estimates. Monitor registration-count queries as data grows and review existing account, status and event indexes before adding new ones.

G11 ReserveIT

Repository:
https://github.com/sev-villanueva/ReserveIT-Classroom-and-Material-Reservation-System-

Branch: main

Database/schema: ReserveIT / public

Result: Healthy | Priority: Follow-up action

Accounts and health

Inspection: reserveit_reporter (app_readonly_grp). Query: reserveit_app (app_runtime_grp). Maintenance: reserveit_owner (object owner).

Local snapshot: 9062 kB; 8 tables, 19 indexes, 0 views. Other sessions visible: 0; visible blocked sessions: 0. Database deadlocks counter: 0; checksum failures counter: 0. Autovacuum is on.

Selected table public.reservation: estimated live rows 8, dead rows 8, modifications since analyze 15.

Logs and maintenance

ANALYZE and ordinary VACUUM on public.reservation; 2026-11-07

ANALYZE and VACUUM completed. Related tables with inaccurate row estimates were also analyzed where permitted. The command output is saved in the project folder.

Query plan and interpretation

Source: main.py:38; staff dashboard status counts

Exact SELECT and first EXPLAIN: explain.sql/txt. Runtime evidence: plan-before.txt and plan-after.txt.

Observed plan after the last authorized action

```
Aggregate: estimated 5, actual 5.0, loops 1
Seq Scan on reservation: estimated 8, actual 8.0, loops 1
```

Execution time before: 13.413 ms; after: 0.124 ms.

Recommendation and limits

The status-count query reads all reservation rows, so a sequential scan is reasonable. Updated statistics now match the small sample; another status index is not justified.

Shared logs and interpretation

Read-only filesystem review using the authorized local Windows account; restricted database accounts could not inspect log_directory.

Available shared server log files dated 2026-11-06 and 2026-11-07. Severity lines only; SQL statements and parameter details excluded.

Configured log_line_prefix is %t only; shared lines do not identify database/account reliably. Do not assign a shared event to a project without its separate execution record.

Category	Observed matching severity lines	Meaning
locking	0	No matching deadlock/timeout severity line in this selected sample.
reliability	0	No matching PANIC/out-of-space/corruption severity line in this selected sample.
connection	62	Includes earlier failed authentication and intended cross-database rejections.
permission	152	Includes deliberate restriction tests and current log-settings denials.
constraint	10	Includes rejected invalid or referentially inconsistent test writes.
schema	2	Historical missing schema/import-order failures; not proof of a current outage.

The totals count matching log lines, not separate incidents. Some entries come from planned permission and constraint tests. File details and examples are in shared-log-review.json.

postgresql-2026-11-06_000000.log: 2026-11-06 00:29:07 CST FATAL: password authentication failed for user "pawpaw_app"

postgresql-2026-11-06_000000.log: 2026-11-06 00:37:46 CST ERROR: schema "dropboxed" does not exist

postgresql-2026-11-06_000000.log: 2026-11-06 01:22:43 CST FATAL: pg_hba.conf rejects connection for host "::1", user "event_app", database "dropboxed_db", no encryption

postgresql-2026-11-06_222517.log: 2026-11-06 22:57:52 CST ERROR: there is no unique constraint matching given keys for referenced table "staff"

Include the database, user and process ID in future log entries to make troubleshooting easier. Keep monitoring privileges separate from normal application accounts.

Testing limitations

Only the inspection session was visible in each successful check. These checks do not cover activity at other times, and reporting accounts may not see details of other users sessions.

Maintenance used the existing database owners or accounts with MAINTAIN. VACUUM ran outside a transaction with short timeouts. No roles, indexes or server settings were changed.

The tables were small, and some were empty. Query timings are individual observations affected by caching and scheduling, so they do not prove a speed improvement. New indexes should be considered only after testing with realistic data and reviewing their storage and write costs.

Recommendations and feedback

Priority	Action
Immediate action	G01/G03: securely configure the intended Activity 4 account credentials, resolve earlier access-control evidence gaps, then rerun inspection. Do not reset passwords or databases without owner direction.
Follow-up action	G05/G08: request scoped maintenance access. Other projects: review remaining statistics/dead rows after bulk tests and monitor representative workload. Improve shared log attribution through owner review.
No urgent incident established	The visible snapshots and selected log patterns do not establish a current lock, corruption or resource-exhaustion incident. This is limited evidence, not an all-clear guarantee.

Feedback drafts are saved in the project folders. Activity 5 feedback has not been sent.

AI usage declaration

AI was used to prepare and run SQL checks, interpret the results and help write this report. The results were checked against the saved output and PostgreSQL documentation.

Technical references

PostgreSQL 18 VACUUM: https://www.postgresql.org/docs/18/sql-vacuum.html

PostgreSQL 18 monitoring statistics and session visibility: https://www.postgresql.org/docs/18/monitoring-stats.html

PostgreSQL 18 privileges and MAINTAIN: https://www.postgresql.org/docs/18/ddl-priv.html